Practical 4

Title:
Exploratory Data Analysis on Air Quality Dataset

Problem:
To perform exploratory data analysis on the inbuilt airquality dataset.

Solution:
The dataset is analyzed using summary statistics, boxplots, histograms, and missing value checks. Interquartile range (IQR) is calculated to identify outliers. Missing values are detected and handled using appropriate techniques such as removal or imputation.

```
# Load the inbuilt airquality dataset

data(airquality)


# View first few rows of the dataset

head(airquality)


# Check structure of the dataset

str(airquality)


# Summary statistics of the dataset

summary(airquality)


# Check total missing values in each column

colSums(is.na(airquality))
```

```
# Replace missing values with mean of respective columns

airquality$Ozone[is.na(airquality$Ozone)] <- mean(airquality$Ozone, na.rm =
TRUE)

airquality$Solar.R[is.na(airquality$Solar.R)] <- mean(airquality$Solar.R, na.rm =
TRUE)


# Verify missing values after imputation

colSums(is.na(airquality))


# Boxplot for Ozone

boxplot(airquality$Ozone,

        main = "Boxplot of Ozone",

        ylab = "Ozone Levels",

        col = "lightblue")


# Calculate IQR for Ozone

Q1 <- quantile(airquality$Ozone, 0.25)

Q3 <- quantile(airquality$Ozone, 0.75)

IQR_value <- IQR(airquality$Ozone)


Q1
```

Q3

```
IQR_value


# Histogram for Ozone
hist(airquality$Ozone,
    main = "Histogram of Ozone",
    xlab = "Ozone Levels",
    col = "lightgreen",
    breaks = 10)
boxplot(airquality$Temp,
       main = "Boxplot of Temperature",
       ylab = "Temperature (F)",
       col = "orange")


# View cleaned dataset
summary(airquality)
```

Expected Outcome:
The program should provide statistical summaries, visualizations, identify outliers, handle missing values, and interpret the overall data distribution.

Practical 5:

Use bar plot and identify the difference between bar plot and histograms. Conclude the appropriate use of bar plot and histogram.

R code for creating bar chart:

Loading dataset:

```
s<-read.csv("C:/Users/shalinik1/OneDrive - KPMG/Desktop/colleges/datasets/store.csv")
```

loading library for creating a summary table:

```
library(dplyr)
```

#regionwise sum of sales

```
e<-s%>%group_by(Region)%>%summarise(p=sum(Sales))
```

Loading library for creating bar chart

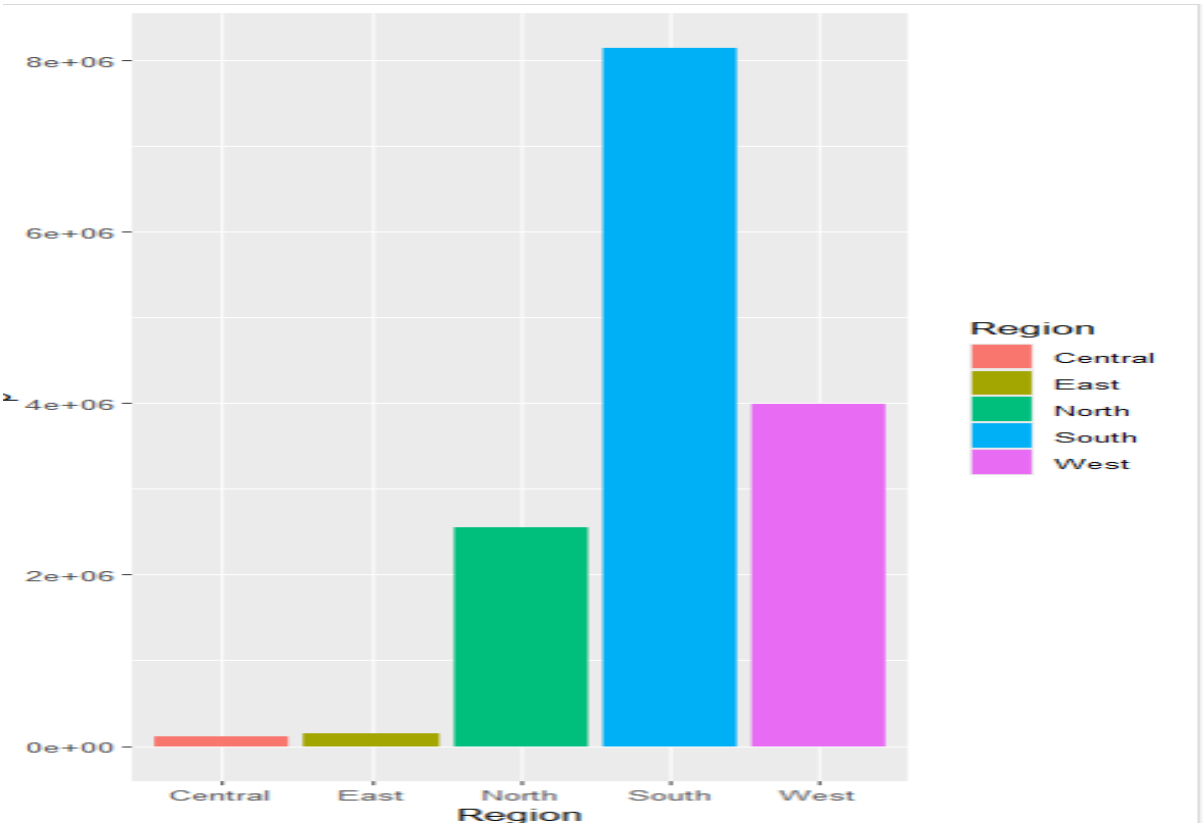
```
library(ggplot2)
```

Bar chart creation:

```
ggplot(e,aes(x=Region,y=p,fill=Region))+geom_bar(stat="identity")
```

Expected Output:

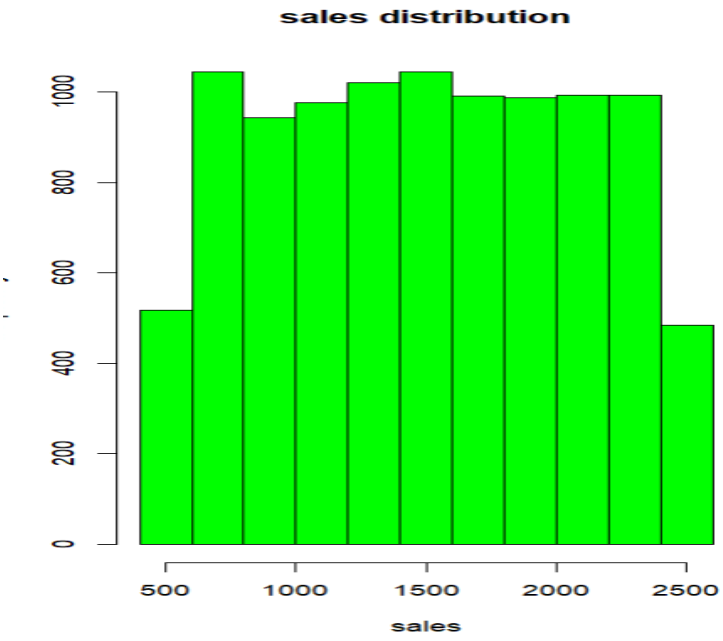
	Region	p
	<chr>	*<int>*
1	Central	117005
2	East	153974
3	North	2552525
4	South	8144192
5	West	3989286



R Code for creating pie chart:

hist(s$Sales,main="sales distribution",xlab="sales",ylab="frequency",col="green",breaks=10)

Expected Output:



Practical 6:

Explore the two-dimensional data:

summary(s)

```
> summary(data)
       X            Company            TypeName             Inches      ScreenResolution       Cpu
 Min.   :   0.0   Length:1303        Length:1303        Min.   :10.10   Length:1303        Length:1303
 1st Qu.: 325.5   Class :character   Class :character   1st Qu.:14.00   Class :character   Class :character
 Median : 651.0   Mode  :character   Mode  :character   Median :15.60   Mode  :character   Mode  :character
 Mean   : 651.0                                         Mean   :15.02
 3rd Qu.: 976.5                                         3rd Qu.:15.60
 Max.   :1302.0                                         Max.   :18.40
     Ram               Memory              Gpu               OpSys              Weight              Price
 Length:1303        Length:1303        Length:1303        Length:1303        Length:1303        Min.   :  9271
 Class :character   Class :character   Class :character   Class :character   Class :character   1st Qu.: 31915
 Mode  :character   Mode  :character   Mode  :character   Mode  :character   Mode  :character   Median : 52055
                                                                                                 Mean   : 59870
                                                                                                 3rd Qu.: 79274
                                                                                                 Max.   :324955
```

cor(data$Price,data$Inches)

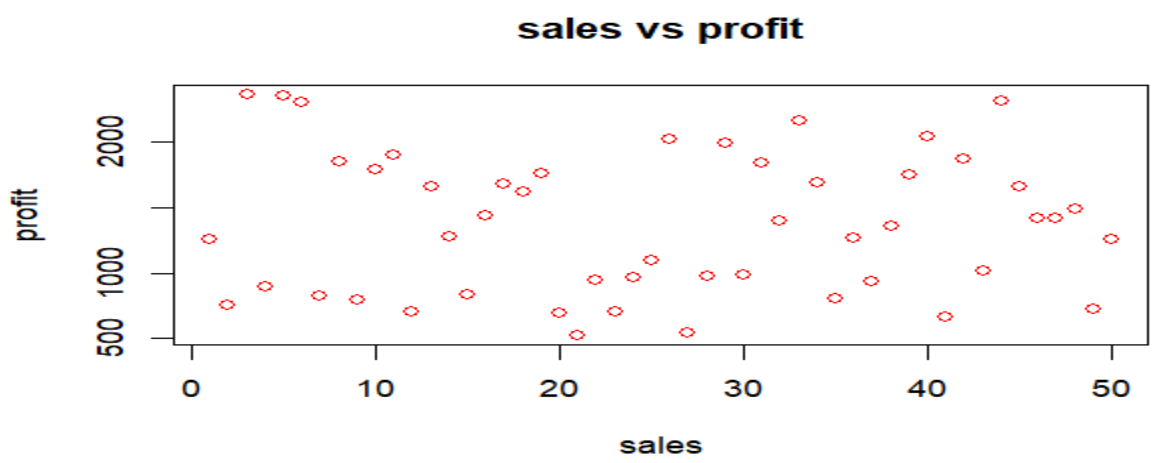
```
> cor(data$Price, data$Inches)
[1] 0.06819667
```

Practical 7:

Scatter plot between two variables

plot(head(s$Sales,50),head(s$profit,50),xlab="sales",ylab="profit",pch=1,main="sales vs profit",col="red")

Expected Output:



Practical 8:

Five number summary in exploratory data analysis

fivenum(s$Profit)

Expected Output:

```
> fivenum(s$Profit)
[1]   25.25  180.00  320.78  525.69 1120.95
```

Practical 9:

Multiple histogram and multiple boxplots, Multiple scatter plots and coloring the graph.

#Multiple Histograms (side by side)

library(ggplot2)

d1<-s$Sales

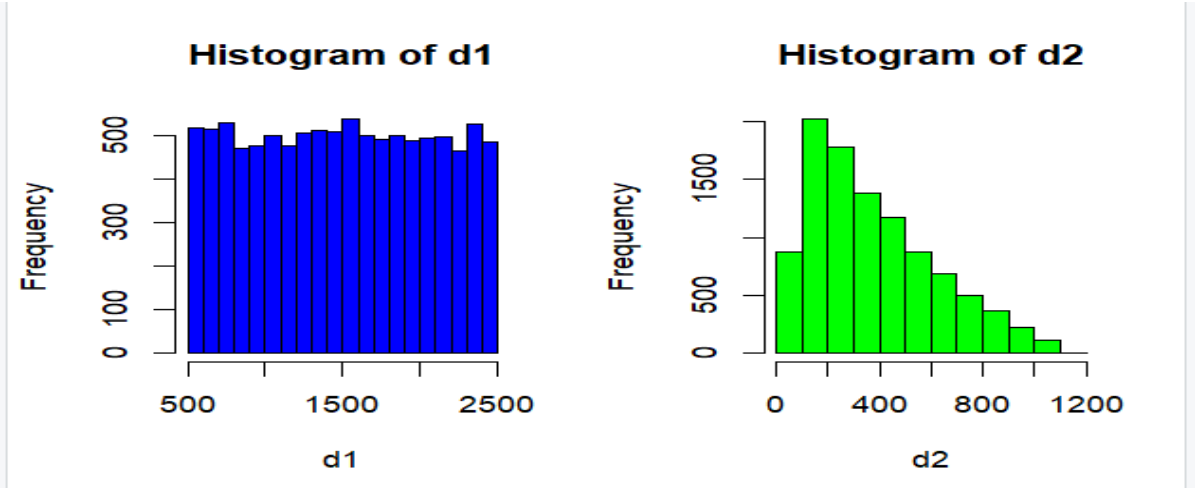
d2<-s$Profit

par(mfrow=c(1,2))

hist(d1,col="blue")

hist(d2,col="green")

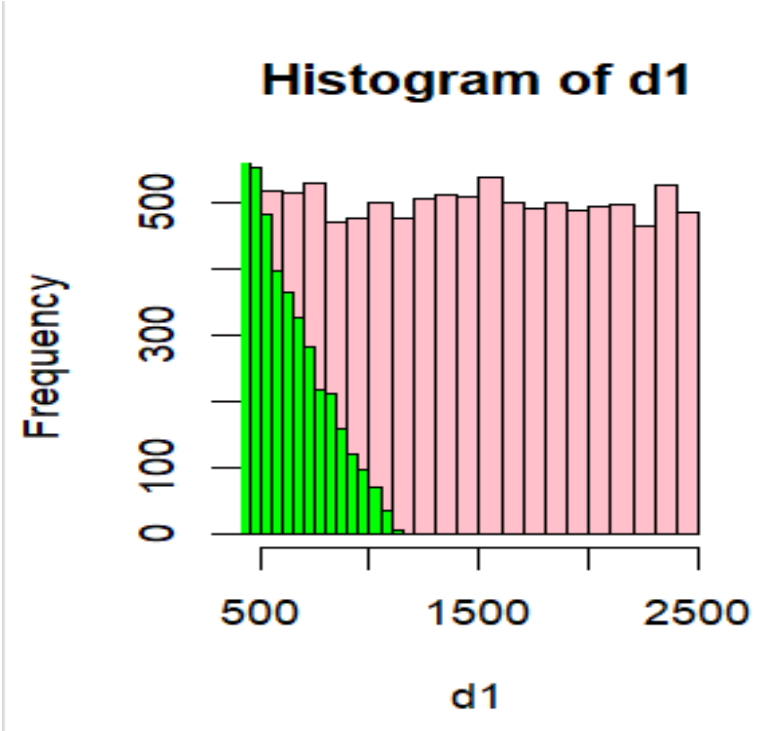
Expected Output:



#Overlapping

```
d1<-s$Sales

d2<-s$Profit

par(mfrow=c(1,2))

hist(d1,col="pink",breaks=15)

hist(d2,col="green",add=TRUE,breaks=20)
```

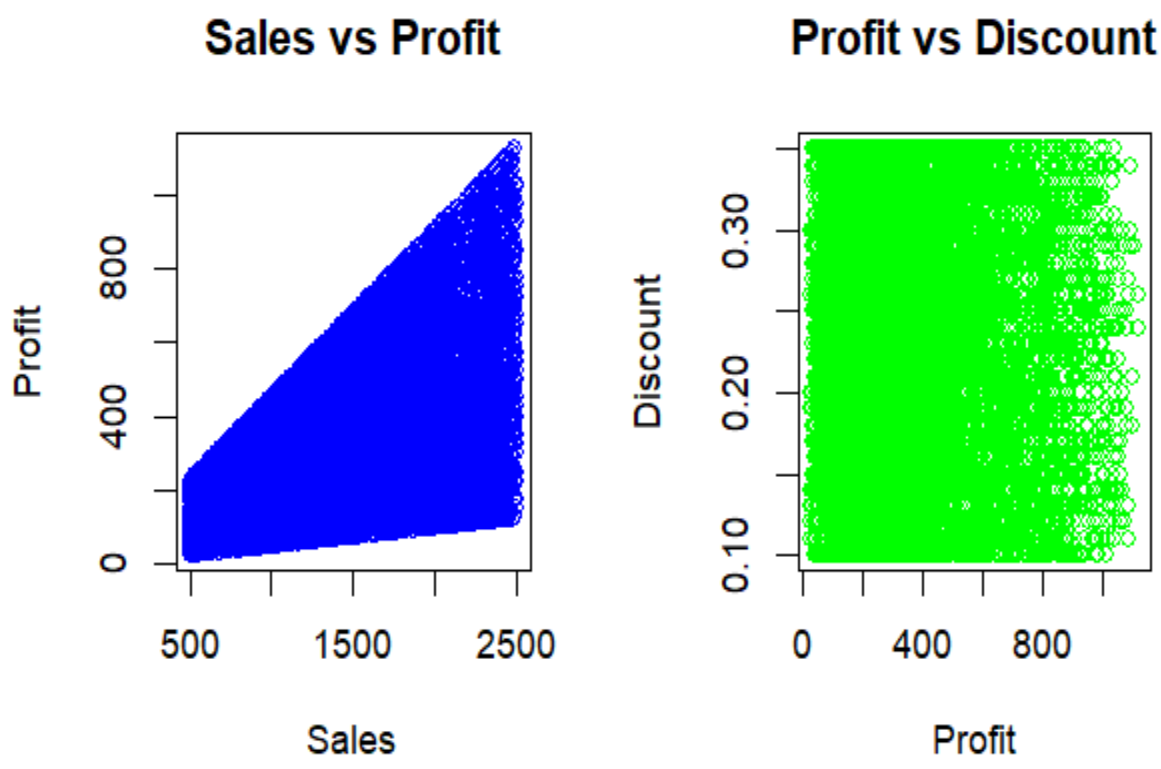
Expected Output:




```
# Multiple scatterplots with titles and axis labels

d1 <- s$Sales

d2 <- s$Profit

d3 <- s$Discount


# Set up the plotting area to have 1 row and 2 columns

par(mfrow = c(1, 2))


# First scatter plot: Sales vs Profit

plot(d1, d2, col = "blue",

     main = "Sales vs Profit",

     xlab = "Sales",

     ylab = "Profit")
```

```
# Second scatter plot: Profit vs Discount
plot(d2, d3, col = "green",
    main = "Profit vs Discount",
    xlab = "Profit",
    ylab = "Discount")
```

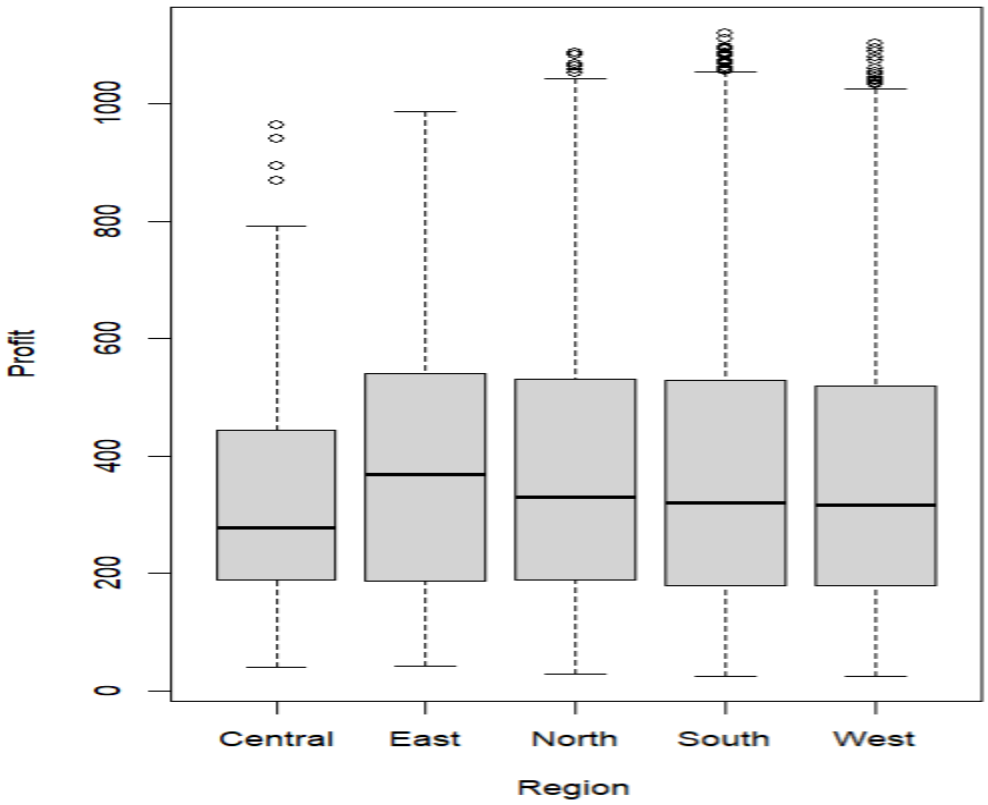
Expected Output:



#Multiple Box plot

```
boxplot(Profit~Region,data=s)
```

Expected Output:



Practical 10: Lattice system in R environment and Graphical window in R and its uses

Use of Lattice in R

The Lattice package in R is primarily used for:

1. Creating Complex Multivariate Plots: It allows for the creation of detailed plots that can display relationships between multiple variables.
2. Conditioning: You can create plots that are divided into panels based on the values of one or more conditioning variables, making it easier to compare subsets of data.
3. Customization: Lattice provides extensive options to customize the appearance of plots, including themes and graphical parameters.

Use of Graphical Window in R

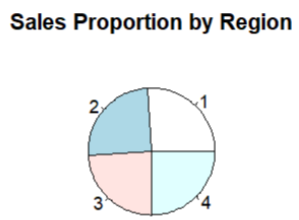
The graphical window in R is used for:

1. Displaying Plots: It opens a window to display various types of plots and graphical outputs.
2. Interactive Graphics: Allows for interactive manipulation of plots, such as adding points or lines to an existing plot.
3. Saving Plots: You can save the plots displayed in the graphical window to files for later use or sharing.

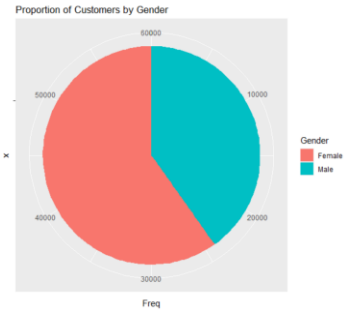
 # Create a pie chart showing the proportion of sales by region using lattice.

```
library(lattice)

sales$total_sales <- sales$price * sales$quantity

region_sales <- aggregate(total_sales ~ Region, data=sales, sum)

pie(region_sales$total_sales, labels=region_sales$region, main="Sales Proportion by Region")
```

Expected Output:



 # Create a pie chart showing the proportion of customers by gender.

```
windows()

gender_counts <- table(customer$gender)

gender_df <- as.data.frame(gender_counts)

ggplot(gender_df, aes(x="", y=Freq, fill=Var1)) +geom_bar(stat="identity", width=1) +coord_polar("y") +

  labs(title="Proportion of Customers by Gender", fill="Gender")
```

Expected Output: